

XMBA 200S: Data and Decisions

Chapter 24: Building Regression Models

Instructor: Reed Walker

Course overview
Part 4: Building regression models

- Blocks 3 and 4
- You want to build a model to forecast outcomes from data you have
- What's the best model you can build?
- This class: when should you add variables to your model?
  - Next block: categorical (0/1) explanatory variables and interactions

Additional variables: what is the trade-off?

Goal of this class: highlight the power of statistical models, but also to illustrate potential risks

Why not just add variables?

- More variables means a higher R-squared
- Explain more variation
- So much data available now – shouldn't we always just use whatever we can?

Additional variables: what is the trade-off?

Additional variables: the risks

- Does your model make sense? Can you explain it to customers or clients?
- Testing many hypotheses (i.e. including many explanatory variables) increases likelihood of spurious correlations
- Sometimes have more explanatory variables than observations… can’t estimate a slope for every variable
- “Overfitting” can lead to poor predictive ability

Outline

- Building a multivariate regression model: pricing Meridian's used gear
- Collinearity
- Automated Model Selection / Machine Learning

Meridian Renewed: what is a used tent worth?

Meridian Renewed buys back used gear and resells it. Want to design an app that quotes a trade-in offer on the spot.

To quote an offer, Meridian has to predict what each item will resell for.

Task: predict resale price from what the app can see. Which variables belong in the model?

Original MSRP ≈ $281	**Age** ≈ 2.7 seasons	**Condition grade** ≈ 6.4 / 10	**Listing photos** ≈ 8.2

Meridian Renewed Data

Response Variable:

Resale price, in $ (104 items sold in the pilot, in sale order)

Explanatory Variables (visible at quote time):

Original MSRP ($)
Age (seasons since release)
Condition grade (1–10, from inspection)
Listing photos (count)

Let’s first consider an SRM with MSRP only.

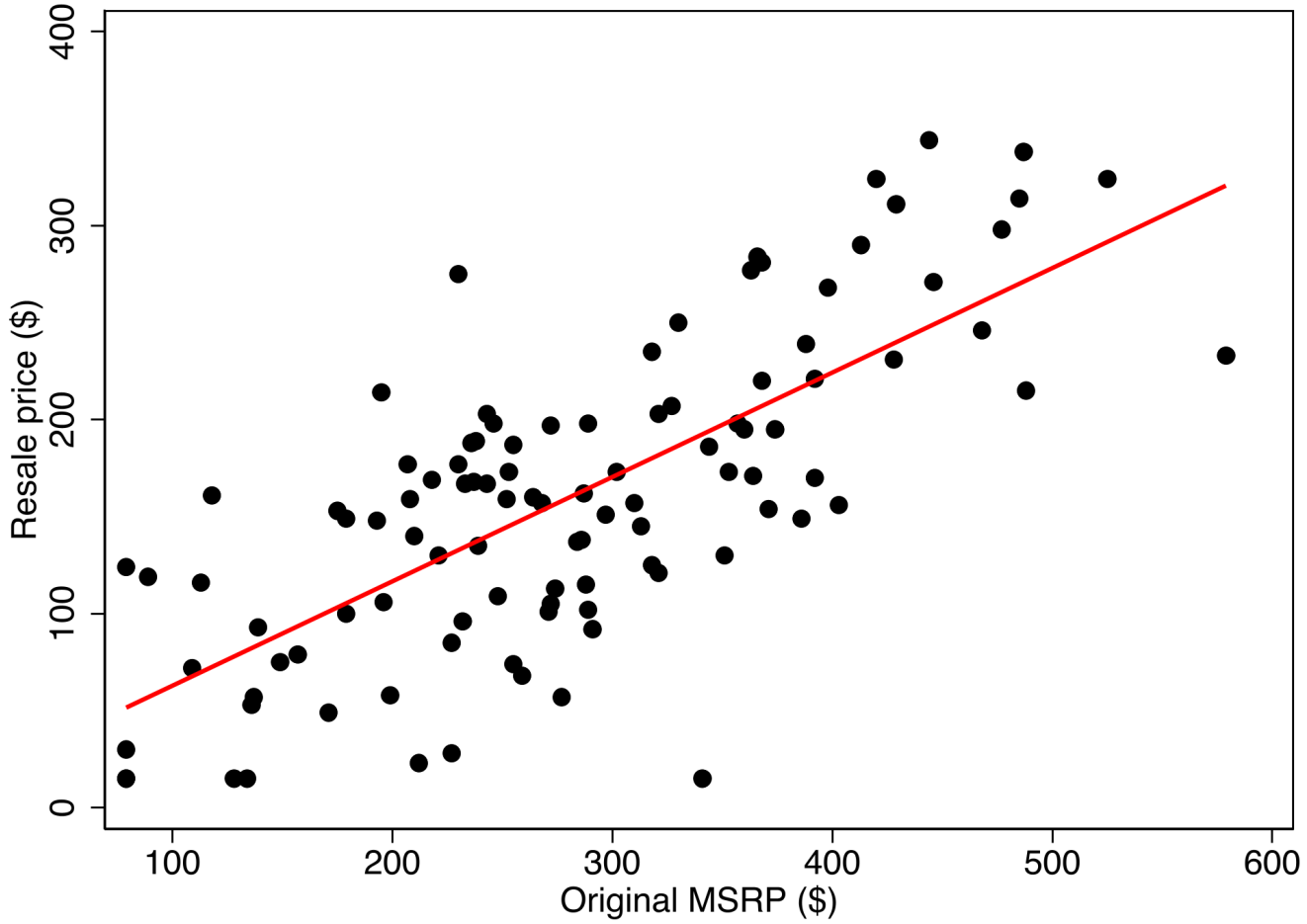
Initial Model (SRM)

Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
resale	104	160.4	79.9	15.0	344.0
msrp	104	281.4	108.5	79.0	579.0
age	104	2.7	1.4	0.7	6.5
condition	104	6.4	1.7	3.0	10.0
photos	104	8.2	2.1	3.0	13.0
holiday	104	0.12	0.32	0	1
clearance	104	0.10	0.30	0	1

Initial Model (SRM)

Scatterplot and Fitted Line



Initial Model (SRM)

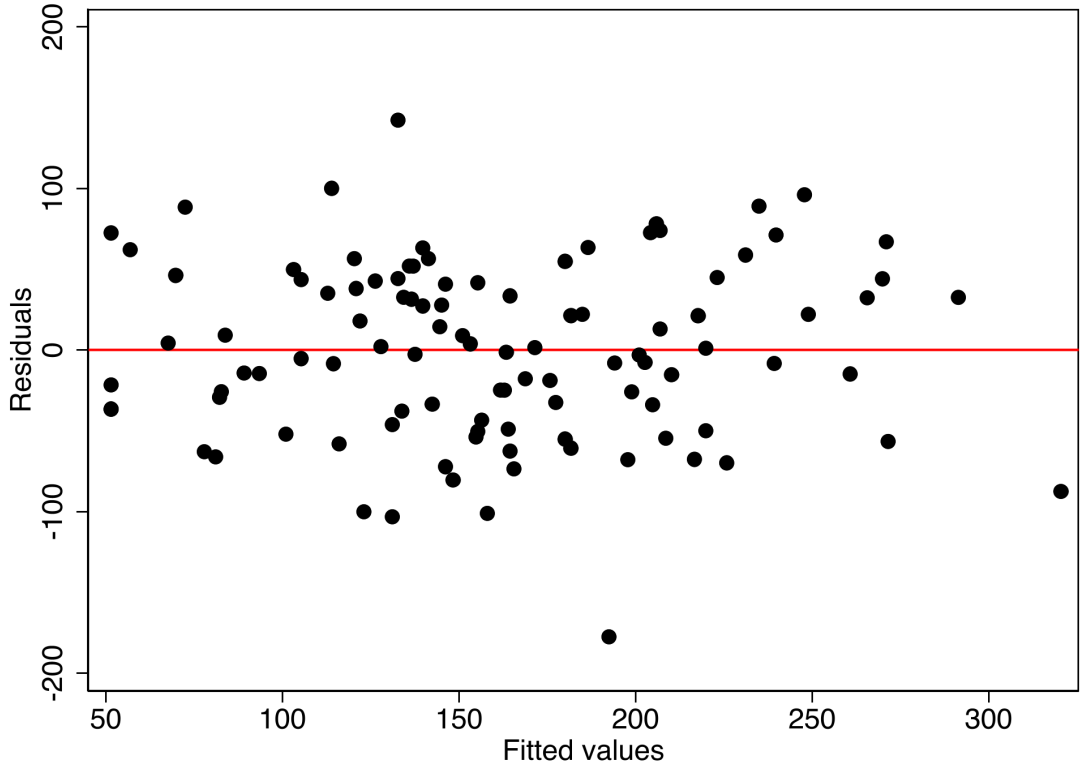
Regression Results

```
      Source |       SS           df       MS      Number of obs   =       104
-------------+----------------------------------   F(1, 102)       =    116.62
       Model |  350838.216         1  350838.216   Prob > F        =    0.0000
    Residual |  306854.822       102  3008.38061   R-squared       =    0.5334
-------------+----------------------------------   Adj R-squared   =    0.5289
       Total |  657693.038       103  6385.36931   Root MSE        =    54.849

------------------------------------------------------------------------------
      resale |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        msrp |   .5379187   .0498115    10.80   0.000     .4391178    .6367195
       _cons |   9.021117   15.01446     0.60   0.549    -20.75999    38.80222
------------------------------------------------------------------------------
```

Initial Model (SRM)

Residual Plots



No pattern against fitted values or sale order.

Expanded Model (MRM)

Add age of product to model

```
      Source |       SS           df       MS      Number of obs   =       104
-------------+----------------------------------   F(2, 101)       =    141.40
       Model |   484618.85         2  242309.425   Prob > F        =    0.0000
    Residual |  173074.189       101  1713.60583   R-squared       =    0.7368
-------------+----------------------------------   Adj R-squared   =    0.7316
       Total |  657693.038       103  6385.36931   Root MSE        =    41.396

------------------------------------------------------------------------------
      resale |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        msrp |   .5080907   .0377453    13.46   0.000     .4332142    .5829672
         age |  -25.96858   2.939051    -8.84   0.000    -31.79887   -20.1383
       _cons |   88.75409    14.4859     6.13   0.000     60.01796    117.4902
------------------------------------------------------------------------------
```

Expanded Model (MRM)

Add condition and photos

```
      Source |       SS           df       MS      Number of obs   =       104
-------------+----------------------------------   F(4, 99)        =     76.29
       Model |  496592.991         4  124148.248   Prob > F        =    0.0000
    Residual |  161100.047        99   1627.2732   R-squared       =    0.7551
-------------+----------------------------------   Adj R-squared   =    0.7452
       Total |  657693.038       103  6385.36931   Root MSE        =    40.339

------------------------------------------------------------------------------
      resale |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        msrp |   .4917755   .0374229    13.14   0.000     .4175204    .5660306
         age |  -12.59189   5.753658    -2.19   0.031    -24.00839   -1.175382
   condition |    14.9867   6.482981     2.31   0.023     2.123058    27.85034
      photos |  -1.931672   3.927875    -0.49   0.624    -9.725429    5.862085
       _cons |  -23.24905    45.0035    -0.52   0.607    -112.5458    66.04765
------------------------------------------------------------------------------
```

Age slope half the size and standard error doubles.
R-squared hasn't moved much. What is going on?

Expanded Model (MRM)

Correlation Matrix

	resale	msrp	age	condit~n	photos
resale	1.0000				
msrp	0.7304	1.0000			
age	-0.5145	-0.0894	1.0000		
condition	0.5710	0.1643	-0.8667	1.0000	
photos	0.5053	0.1735	-0.7496	0.8736	1.0000

age, condition and photos are highly correlated

- gear wears with age
- merchandise department photographs items that look better.

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Collinearity: Highly correlated explanatory variables

Definition and Diagnosis

- Collinearity is the situation in which two or more of the explanatory variables are highly correlated.
- Some correlation between explanatory variables is inevitable.
- However, when your explanatory variables are highly correlated this decreases the precision of the estimates.
- In particular, standard errors tend to increase, and point estimates may vary across specifications.

Collinearity

An Example of Perfect Collinearity

```
note: release_year omitted because of collinearity
      Source |       SS           df       MS      Number of obs   =       104
-------------+----------------------------------   F(1, 102)       =     36.73
       Model |  174114.125         1  174114.125   Prob > F        =    0.0000
    Residual |  483578.914       102  4740.96974   R-squared       =    0.2647
-------------+----------------------------------   Adj R-squared   =    0.2575
       Total |  657693.038       103  6385.36931   Root MSE        =    68.855

------------------------------------------------------------------------------
      resale |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         age |  -29.50697   4.869016    -6.06   0.000    -39.16464    -19.8493
release_year |          0  (omitted)
       _cons |   241.4629   14.98322    16.12   0.000     211.7438    271.1821
------------------------------------------------------------------------------
```

Dealing with Collinearity

Learn to recognize the presence of collinearity:

- High correlation between your explanatory variables,
- R-squared doesn't change much as you add a variable,
- Standard errors of slopes increase a lot when you add a variable,
- Slope estimates change dramatically when you add a variable
- Large F-statistic but many small t-statistics.

What do do if there is clear collinearity?

- Do you have a good reason to keep multiple variables in the model?
- If not, consider dropping variables from the model, working one at a time

Refining the Model

Drop photos

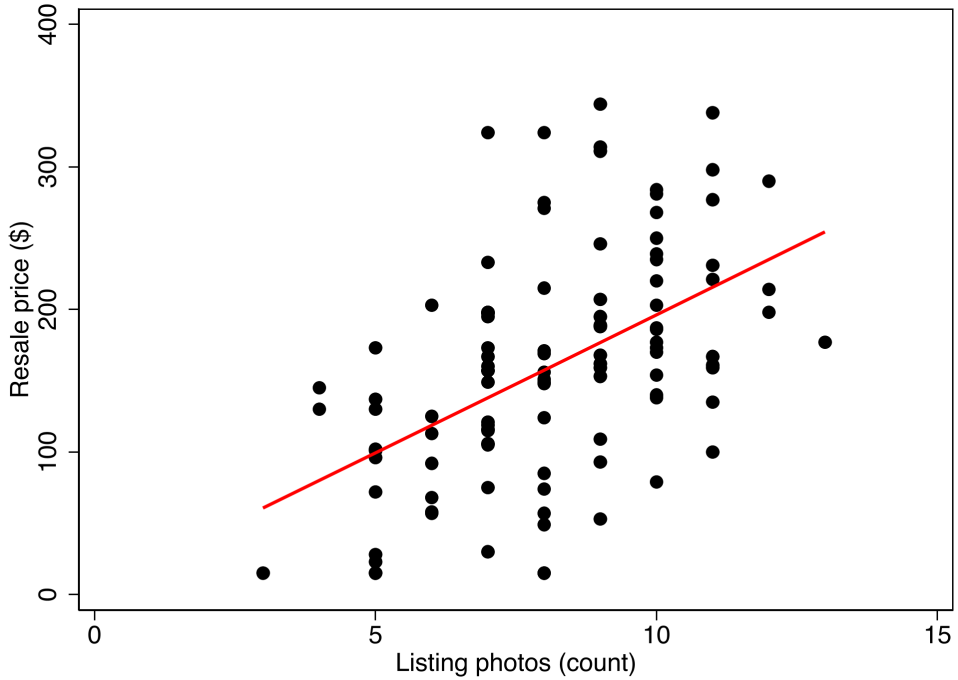
```
      Source |       SS           df       MS      Number of obs   =       104
-------------+----------------------------------   F(3, 100)       =    102.42
       Model |  496199.431         3   165399.81   Prob > F        =    0.0000
    Residual |  161493.608       100  1614.93608   R-squared       =    0.7545
-------------+----------------------------------   Adj R-squared   =    0.7471
       Total |  657693.038       103  6385.36931   Root MSE        =    40.186

------------------------------------------------------------------------------
      resale |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        msrp |    .490681   .0372148    13.19   0.000     .4168479     .564514
         age |  -12.66178   5.730057    -2.21   0.029    -24.03005   -1.293512
   condition |   12.85361   4.799959     2.68   0.009     3.330625    22.37659
       _cons |  -24.89291   44.70874    -0.56   0.579    -113.5938    63.80796
------------------------------------------------------------------------------
```

R^2 0.755 → 0.754. Every slope is now statistically significant.

Refining the Model

But photos looked strong on its own



Photos alone:
slope 19.4, t = 5.9, R^2 = 0.26

Photos in the full model:
slope -1.9, t = -0.5

Merchandising walks by: "so should we require more photos?"

What is the data generating process that created scatterplot?

- Photos -> Higher Resale Price?
- Quality -> Photos -> Higher Retail Price?

This is a causal question, and causal questions require…

What is the “Best” Model?

Typically combines some intuition with model testing:

1. What are candidate hypotheses?

2. Does the model have predictive power with all of these explanatory variables? [F-stat!]

3. Next, does each explanatory variable have statistical significance? [t-stat!]
   - If “yes” – great
   - If “no” – maybe exclude variable or maybe not
     - By excluding you are forcing the relationship to be 0
     - By including, you may be adding “noise” to model and giving up parsimony. Judgement call.

4. If excluding a variable – do so, 1 at a time. Start back at #1

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Automated Model Selection / Machine Learning

- The previous model building “algorithm” seems a bit ad hoc
  - Are there more formal ways of picking the ”right” model?
- Answer: yes!
  - **Machine Learning:** automated model building procedures
- These methods come with their own issues:
  - Overfitting and Spurious Correlations
  - They cannot account for common sense
- And some of these *problems* also have *solutions*:
  - Cross-validation + Training Sets + Testing Sets

Can random numbers predict resale prices?

Source	SS	df	MS	Number of obs	=	104
				F(24, 79)	=	14.50
Model	536042.53	24	22335.1054	Prob > F	=	0.0000
Residual	121650.509	79	1539.87986	R-squared	=	0.8150
				Adj R-squared	=	0.7588
Total	657693.038	103	6385.36931	Root MSE	=	39.241

resale	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
msrp	.4852923	.0412725	11.76	0.000	.4031415	.5674432
age	-9.746318	6.681129	-1.46	0.149	-23.04477	3.552137
condition	17.62373	7.414411	2.38	0.020	2.865715	32.38175
photos	-1.518782	4.423161	-0.34	0.732	-10.32286	7.285299
junk1	1.838262	4.253364	0.43	0.667	-6.627848	10.30437
junk2	-5.295097	4.355675	-1.22	0.228	-13.96485	3.374657
junk3	8.55937	3.830136	2.23	0.028	.9356753	16.18306
junk4	2.743754	4.30958	0.64	0.526	-5.83425	11.32176
junk5	-.8338628	4.839582	-0.17	0.864	-10.46681	8.799084
junk6	.4420429	3.90302	0.11	0.910	-7.326725	8.210811
junk7	-3.392613	4.280197	-0.79	0.430	-11.91213	5.126905
junk8	-11.82178	4.499871	-2.63	0.010	-20.77855	-2.865015
junk9	-2.268013	3.830466	-0.59	0.555	-9.892365	5.356339
junk10	5.991318	4.284734	1.40	0.166	-2.537231	14.51987
junk11	-1.752847	4.216508	-0.42	0.679	-10.1456	6.639902
junk12	-3.596711	4.801952	-0.75	0.456	-13.15476	5.961336
junk13	-6.787535	4.462949	-1.52	0.132	-15.67081	2.095742
junk14	2.159993	4.356063	0.50	0.621	-6.510533	10.83052
junk15	-2.160289	4.288746	-0.50	0.616	-10.69682	6.376245
junk16	-2.727019	4.491879	-0.61	0.546	-11.66788	6.213843
junk17	-.8297169	4.912921	-0.17	0.866	-10.60864	8.949208
junk18	-5.145241	4.1919	-1.23	0.223	-13.48901	3.198526
junk19	1.627877	4.354755	0.37	0.710	-7.040046	10.2958
junk20	7.123744	4.043796	1.76	0.082	-.9252313	15.17272
_cons	-54.02662	52.41049	-1.03	0.306	-158.3471	50.29385

Add 20 random variables: R^2 climbs 0.76 → 0.81, and two of them look significant.

Random Numbers Predict Resale Prices

```
      Source |       SS           df       MS      Number of obs   =       104
-------------+------------------------------       F(1, 102)       =      8.04
       Model |  48034.2095         1  48034.2095    Prob > F        =    0.0055
    Residual |  609658.829       102  5977.04734    R-squared       =    0.0730
-------------+------------------------------       Adj R-squared   =    0.0639
       Total |  657693.038       103  6385.36931    Root MSE        =    77.311

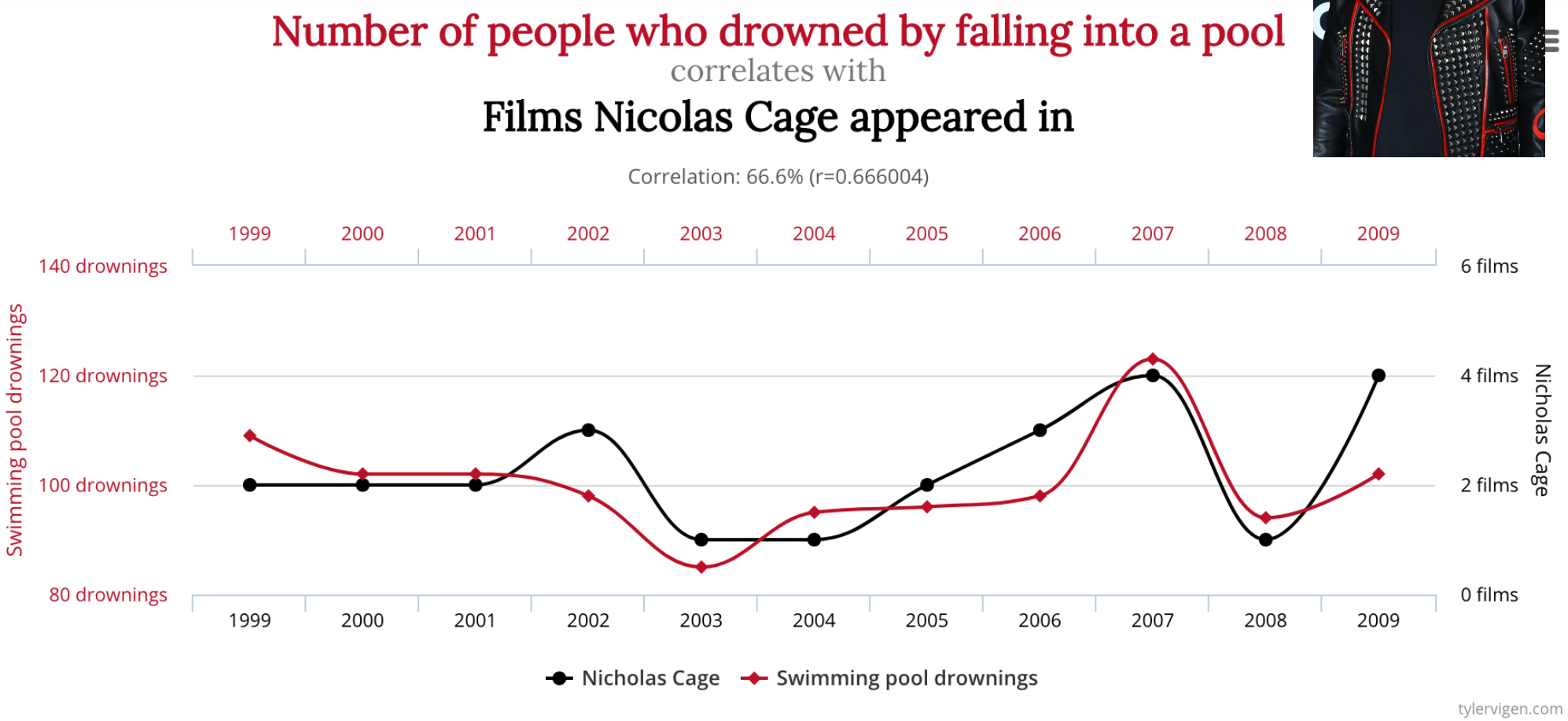
------------------------------------------------------------------------------
      resale |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       junk8 |  -22.89869    8.07753    -2.83   0.006    -38.92043   -6.876943
       _cons |   155.4965   7.776132    20.00   0.000     140.0726    170.9204
------------------------------------------------------------------------------
```

junk8: t = -2.83, p = 0.006.

Should the pricing model include junk8?

Back to Nicolas Cage…





Data sources: Centers for Disease Control & Prevention and Internet Movie Database

Machine Learning and LASSO

- Probably the best-known ML algorithm is LASSO, a.k.a. “Least Absolute Shrinkage and Selection Operator”.
  - Other common algorithms are “decision trees” and “random forests”

- The idea with LASSO is to minimize the sum of squared residuals, just as with our usual regression model.

- However, in LASSO the slopes are subject to a constraint,

$$\sum_{k=1}^{K} |b_k| < \lambda.$$



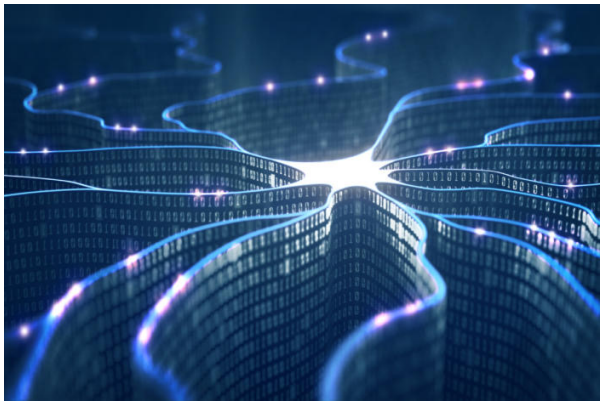
Machine Learning and LASSO

- That is, the sum of the absolute values of all slope estimates can't exceed a "penalty parameter", λ.

- This "penalty" helps avoid spurious correlation by emphasizing only a few highly-predictive explanatory variables.
  - Often when you use LASSO, you get most explanatory variables with zero slope, and a few with statistically significant slopes.

- Great! But what λ should I use? And don't I still run the risk of finding spurious correlations?





Cross Validation

- Prior to using LASSO (or other ML algorithms) we often divide our sample in two parts.

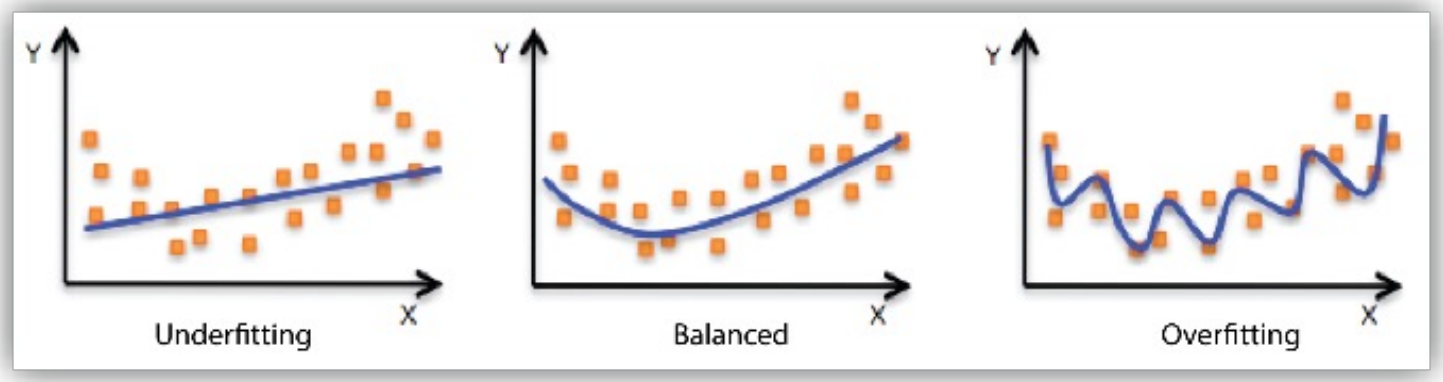


- We estimate the model using the first sample, and a range of different λ's.

- But then we use the second sample to calculate residuals.
  - We calculate residuals for the second sample, using the slopes we estimated with first sample.
  - For each λ, we have a different set of estimates, and thus a different set of residuals.

Cross Validation

- Finally, we pick the λ which yields the smallest sum of squared residuals in the second sample.
  - For too low λ, we say model is “underfitting”, and excluding predictive variables.
  - For too high λ, we say model is “overfitting”, because its including “too many” variables.

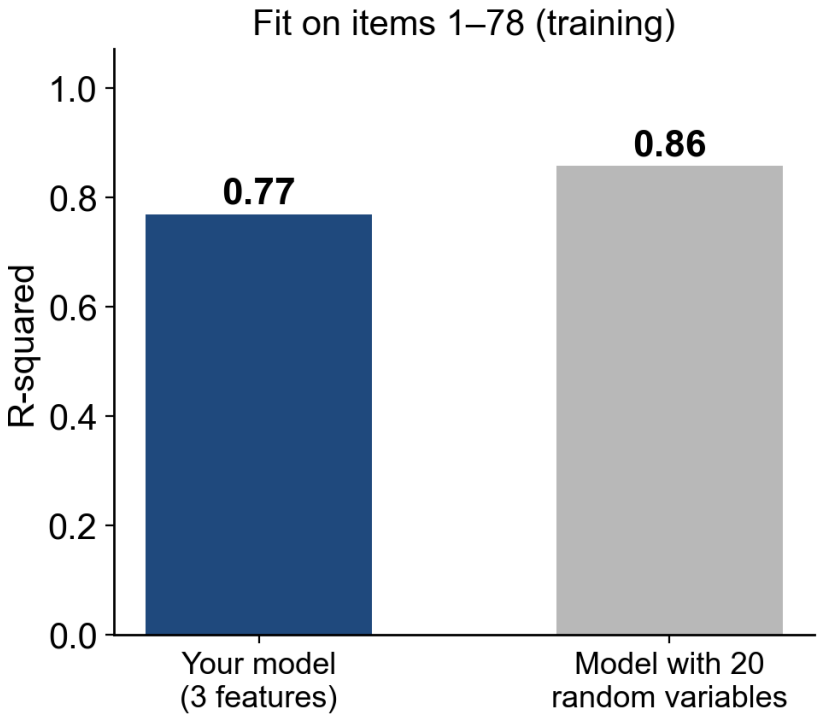


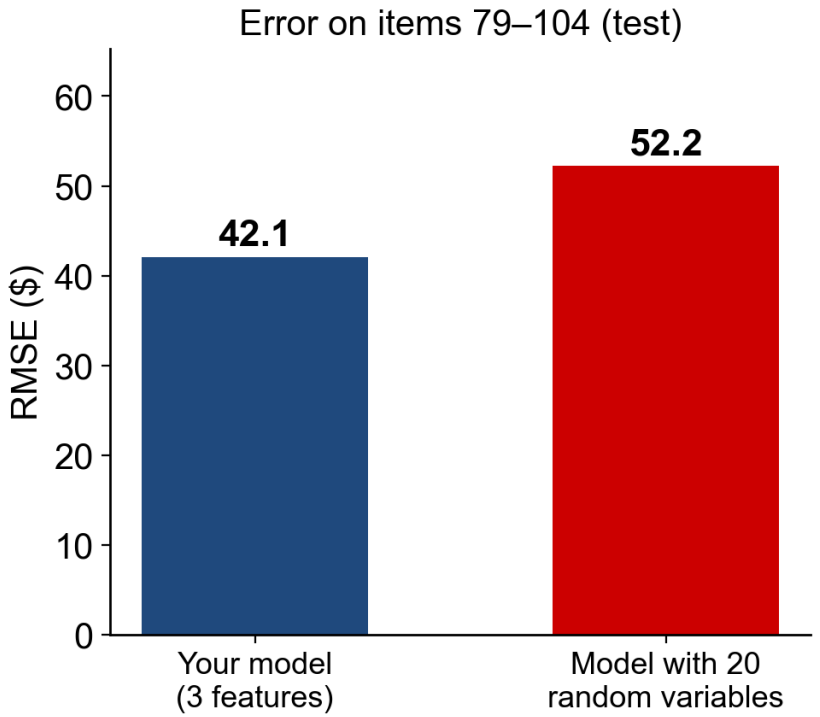
- Cross-validation, particularly when combined with this penalized regression, greatly reduces the risk of spurious correlation.

Cross Validation:
Test the model on data it has never seen

observations 1–78: build the models

79–104: observations used for testing

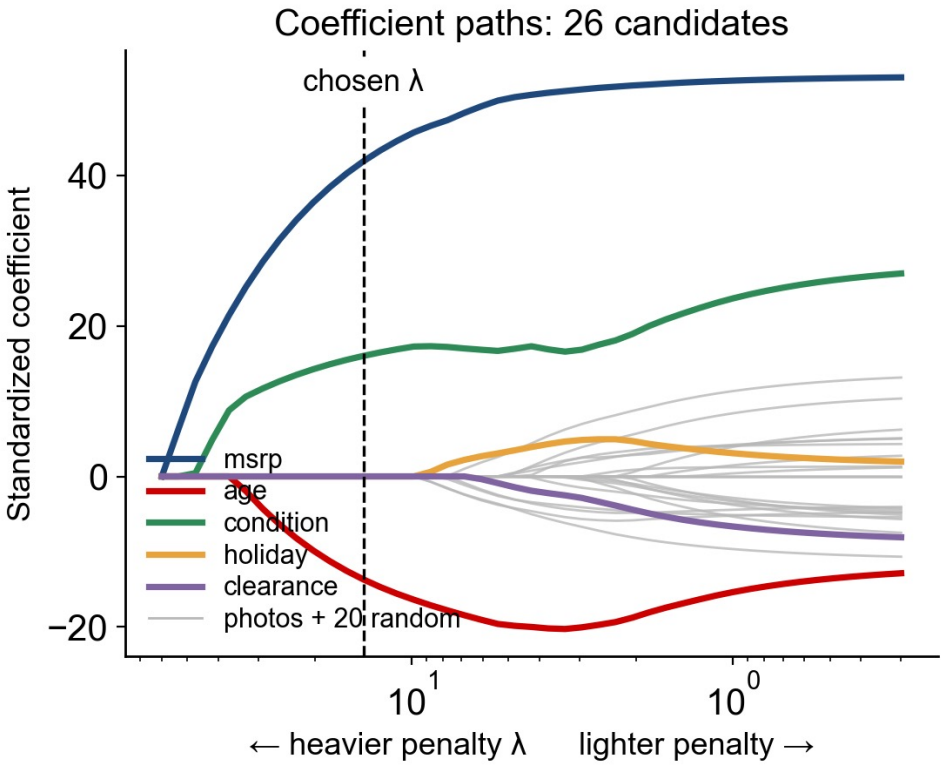


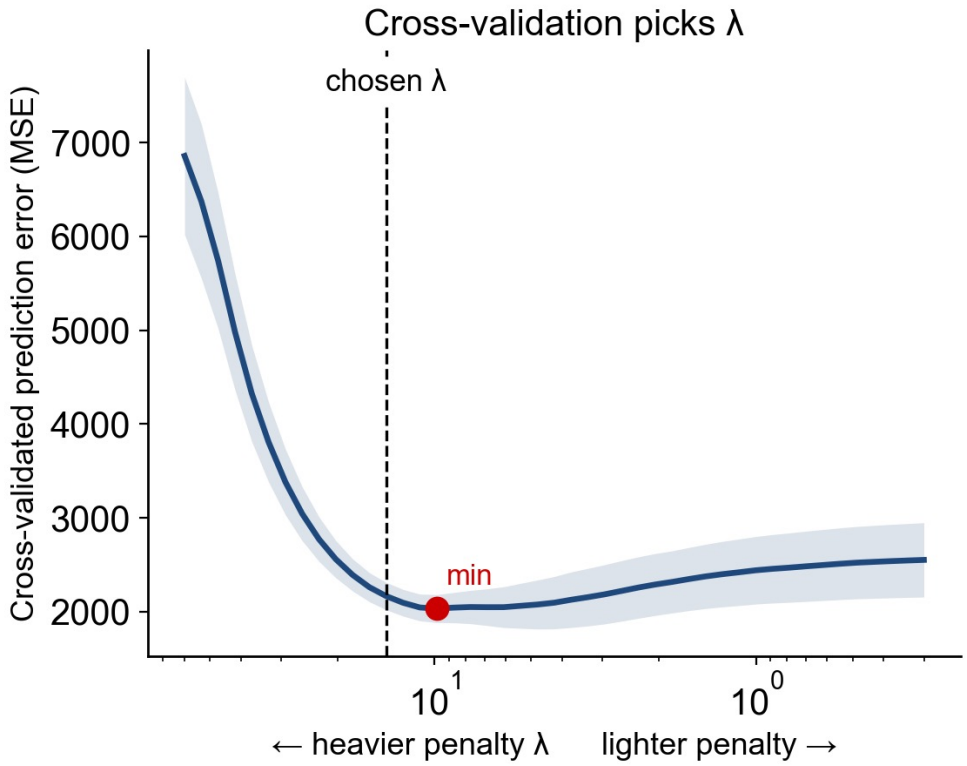


The 20 random variables helped on the training data (higher R^2) and hurt on the test data (higher RMSE)

LASSO on the Meridian Renewed data

26 candidates: 4 features + holiday + clearance + 20 random





What LASSO kept

```
----------------------
             |     active
-------------+----------
        msrp |   .2911079
         age |  -4.638811
   condition |   7.548911
       _cons |    47.6209
----------------------
```

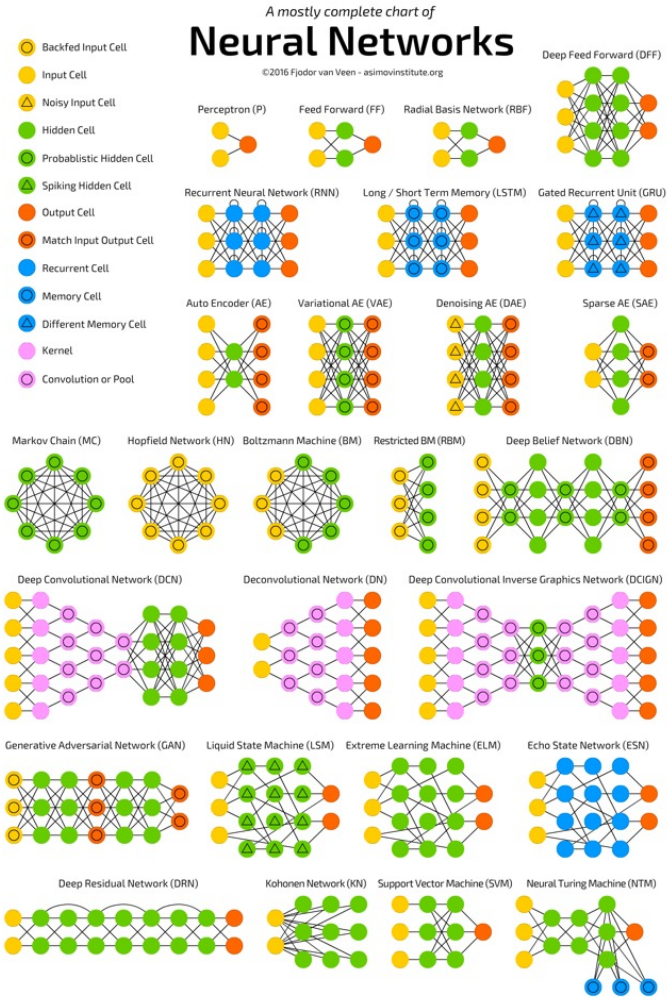
Kept: msrp, age, condition

Dropped: photos, holiday, clearance, and all 20 random variables.

Screening 26 candidate variables took about a second.

Other Flavors of Machine Learning





- Classification/recommendation algorithms: logistic regression, random forests…
- Combination of different ML algorithms can be combined via ensemble methods to get even better predictions

Foundation Models / LLMs

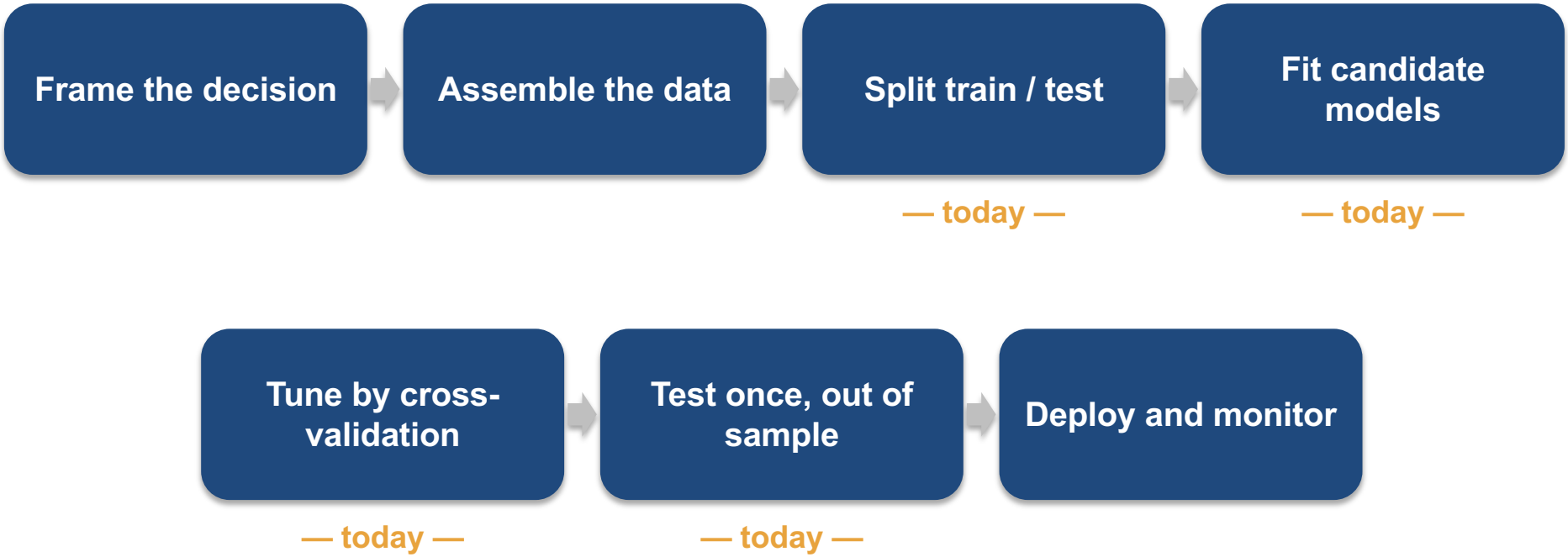
Claude, GPT, Gemini: predictive models, pre-trained on nearly everything

you adapt them with prompts or fine-tuning instead of building from scratch.



Same design recipe: fit flexibly, penalize complexity, judged by out-of-sample error.

The standard machine learning workflow



Steps 3-6 are what we did today.

Monitoring is never finished: performance may drift as the world changes.

Best Practices

- ML with cross-validation is not a panacea
- ML does provide a more objective approach for data mining exercises, while also greatly reducing the risk of spurious correlation.
- Machine learning works best with large datasets and when you have a large number of potential explanatory variables.
- Machine learning can also greatly improve the predictive power of your models, relative to traditional regression models.
- The exact algorithm you use is less important than having a representative sample and asking the right question.

Data Science, Economics, and Lessons Learned…

THE INDUSTRY

Zillow Torched $381 Million Overpaying for Houses. Spectacular.

BY ALEX KIRSHNER NOV 03, 2021 • 1:14 PM



- Not hard to predict the sale price of a home. You did it yesterday!
- Much harder to predict the arbitrage opportunities
- Problem here: **winner's curse** + (possible) **adverse selection**

- Winners Curse: bidders have different private (ex ante) signals about house value but same (ex post) value… winner is bidder with most optimistic view (tend to overestimate and overpay).
- Adverse selection: when sellers have more private information than buyers… you're going to win the crappier houses more often

WHY IS INTERMEDIATING HOUSES SO DIFFICULT? EVIDENCE FROM IBUYERS

Greg Buchak
Gregor Matvos
Tomasz Piskorski
Amit Seru

Working Paper 28252
http://www.nber.org/papers/w28252

ABSTRACT

We study the frictions in dealer-intermediation in residential real estate through the lens of "iBuyers," technology entrants, who purchase and sell residential real estate through online platforms. iBuyers supply liquidity to households by allowing them to avoid a lengthy sale process. They sell houses quickly and earn a 5% spread. Their prices are well explained by a simple hedonic model, consistent with their use of algorithmic pricing. iBuyers choose to intermediate in markets that are liquid and in which automated valuation models have low pricing error. These facts suggest that iBuyers' speedy offers come at the cost of information loss concerning house attributes that are difficult to capture in an algorithm, resulting in adverse selection. We calibrate a dynamic structural search model with adverse selection to understand the economic forces underlying the tradeoffs of dealer intermediation in this market. The model reveals the central tradeoff to intermediating in residential real estate. To provide valuable liquidity service, transactions must be closed quickly. Yet, the intermediary must also be able to price houses precisely to avoid adverse selection, which is difficult to accomplish quickly. Low underlying liquidity exacerbates adverse selection. Our analysis suggests that iBuyers'